

Planck constraints on photon-coupled compact clumps as dark matter: profile-likelihood results

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Abstract

We report the first real-data constraint from this project on the momentum coupling between the cosmic microwave background (CMB) photons and a population of compact, pressureless clumps that make up all of the dark matter. The coupling is expressed through the dimensionless interaction strength u used by the CLASS Boltzmann code, which maps directly onto the clump cross-section per unit mass and hence onto a minimum clump surface density. Profiling the likelihood over the six standard cosmological parameters and all instrumental and foreground nuisance parameters, the Planck 2018 temperature and polarization data give $u < 1.00 \times 10^{-4}$ at 95% confidence, equivalent to $\sigma/M < 3.7 \times 10^{-7} \text{ cm}^2 \text{ g}^{-1}$ and a clump surface density $\Sigma/Q > 2.7 \times 10^6 \text{ g cm}^{-2}$, where Q is an order-unity momentum-transfer efficiency. The independent Planck PR4 (NPIPE) CamSpec high- ℓ likelihood gives a limit 12% weaker, $u < 1.12 \times 10^{-4}$. At zero coupling the pipeline reproduces Planck’s own Λ CDM fit, and the limits agree with a Fisher forecast made before any data were used. These numbers are conditional on a specific and deliberately narrow model, in which the clumps are added to an otherwise standard Λ CDM universe with homogeneous recombination, a smooth diffuse baryon fluid and adiabatic initial conditions. Physics that would plausibly accompany the formation of such clumps, such as small-scale inhomogeneity in the remaining gas or primordial magnetic fields, is not included and could move the limit in either direction. A Bayesian analysis with a flat prior on u gives $u < 1.69 \times 10^{-4}$ ($\Sigma/Q > 1.6 \times 10^6 \text{ g cm}^{-2}$), close to published Bayesian limits. The difference from the profile limit follows from the nearly linear rise of the profile. The coupling is strongly anticorrelated with σ_8 , which shows that the limit comes from the suppression of structure seen through CMB lensing. When only a fraction f_{cl} of the dark matter is in clumps and the rest is collisionless CDM, the permitted coupling grows faster than $1/f_{\text{cl}}$: the profile threshold rises from $u = 1.1 \times 10^{-4}$ at $f_{\text{cl}} = 1$ to 4×10^{-3} at $f_{\text{cl}} = 0.1$ and 0.15 at $f_{\text{cl}} = 0.01$, corresponding to minimum surface densities from 2.4×10^6 down to about $2 \times 10^3 \text{ g cm}^{-2}$. Replacing Thomson scattering off the clumps by uniform, unpolarized re-emission changes these thresholds by a few per cent, except for the smallest fraction at strong coupling.

1 The question

The project asks a conditional question. If a fraction f_{cl} of the dark matter were in compact, pressureless clumps of ordinary matter that exchange momentum with the radiation field through an effective cross-section σ , how large could the cross-section per unit mass σ/M be before the CMB anisotropies record the interaction? Whether such clumps can form, survive and be reconciled with Big Bang nucleosynthesis is a separate question about the premise, which we set aside here. A longer explainer covering the physics from first principles accompanies this report; the present document concentrates on the likelihood analysis and its result.

It is worth being explicit about what kind of statement the answer is. A likelihood analysis with CLASS does not ask which universes could produce the observed sky; it asks which parameter values within one family of dynamical models do so. The family used here is Λ CDM, with its homogeneous recombination, linear perturbations of a smooth baryon fluid, standard atomic and

neutrino physics, and adiabatic, nearly scale-invariant initial conditions, extended by a single new ingredient: a pressureless component that feels a drag force from the radiation. The limit on u is therefore a statement of the form “if the clumps are the only departure from Λ CDM, a coupling larger than this worsens the fit to Planck by more than $\Delta\chi^2 = 2.71$ ”. It is not a model-independent measurement of the opacity of baryonic clumps. Section 6 discusses which of the fixed assumptions a real clump population might violate, and what is known about how relaxing them changes the CMB.

Following [Stadler & Boehm \(2018\)](#) and the interacting dark matter module of CLASS ([Lesgourgues, 2011](#); [Becker et al., 2021](#)), the interaction is described by

$$u = \frac{\sigma}{\sigma_T} \left(\frac{M}{100 \text{ GeV}} \right)^{-1} \iff \frac{\sigma}{M} = \frac{u}{268} \text{ cm}^2 \text{ g}^{-1}, \quad (1)$$

where σ_T is the Thomson cross-section. For optically thick clumps the momentum-transfer cross-section is geometric, $\sigma = Q \pi R^2$, with $Q = 1$ for a perfect absorber and a value of order unity for other reflection laws, so an upper limit on u becomes a lower limit on the clump surface density,

$$\frac{\Sigma}{Q} = \frac{M}{Q \pi R^2} = \frac{268}{u} \text{ g cm}^{-2}. \quad (2)$$

The coupling drags the clumps along with the radiation before recombination, which suppresses the growth of their density perturbations on small scales. The CMB sees this through the weaker gravitational potentials in the damping tail and, most strongly, through the reduced gravitational lensing of the CMB by intervening structure. The photons themselves scatter off the clumps far too rarely to matter at the couplings probed here, which is why the result does not depend on the detailed angular law of the photon-clump interaction. All results in this report assume $f_{\text{cl}} = 1$ and a cross-section independent of temperature.

The same calculation has a history in particle cosmology, where the dark matter is a weakly interacting particle that scatters photons. [Wilkinson et al. \(2014\)](#) obtained the first Planck constraint with a modified version of CLASS, finding $\sigma < 8 \times 10^{-31} (m/\text{GeV}) \text{ cm}^2$ at 68% confidence, or $u \lesssim 1.2 \times 10^{-4}$, from the 2013 data. [Stadler & Boehm \(2018\)](#) repeated the analysis for the 2015 release with a more complete treatment of the photon Boltzmann hierarchy, and [Zhou et al. \(2022\)](#) used the 2018 release in a model in which the dark matter mass is also free. The step from particles to macroscopic objects was taken by [Jacobs et al. \(2015\)](#), who proposed that the dark matter might consist of objects of gram-scale mass or more with geometric cross-sections, and who applied the bound of [Wilkinson et al. \(2014\)](#) to them as $\sigma_X/M_X < 4.5 \times 10^{-7} \text{ cm}^2 \text{ g}^{-1}$. Later studies of such objects concentrated on other effects: [Caloni et al. \(2021\)](#) found that for objects made of ordinary matter the strongest cosmological bound comes from the capture of baryons between nucleosynthesis and recombination, and a recent compilation of constraints as a function of the fraction of dark matter in macroscopic form ([Picker et al., 2026](#)) leaves out the cosmological scattering bounds altogether as a topic to revisit. As far as we are aware, the present work is the first dedicated likelihood analysis of the photon coupling for optically thick clumps, including a check that the bound does not depend on the angular law of the photon-clump interaction.

2 Data and method

2.1 Likelihoods

We use the Planck likelihoods as distributed through the Cobaya framework ([Torrado & Lewis, 2021](#)). Every combination includes the 2018 low- ℓ temperature likelihood (Commander, $2 \leq \ell < 30$) and the low- ℓ E-mode polarization likelihood (SimAll), which together constrain the optical depth to reionization ([Planck Collaboration, 2020a](#)). To these we add one of three high- ℓ temperature and polarization likelihoods ($\ell \geq 30$, TT, TE and EE):

- the 2018 PLIK likelihood, Planck’s baseline, with 21 foreground and calibration nuisance parameters (Planck Collaboration, 2020a);
- PLIK-lite, a compressed version of the same data in which the foregrounds have been marginalised in advance, leaving a single calibration parameter;
- the PR4 (NPIPE) CamSpec likelihood (Rosenberg et al., 2022), built from the final Planck processing with a different treatment of foregrounds and sky masks, and 9 nuisance parameters.

We do not yet include the Planck lensing reconstruction likelihood, which measures the lensing potential directly from the four-point function of the maps. The lensing information used here is only that contained in the smoothing of the acoustic peaks in the power spectra.

2.2 Theory calculation

The CMB spectra are computed with CLASS version 3.4.0, in which the clumps are the interacting dark matter species with density $\omega_{\text{idm}} = f_{\text{cl}} \omega_{\text{dm}}$, a temperature-independent cross-section and a particle mass of 10^{27} eV, which makes their thermal velocities negligible. Early tests showed that the default numerical settings mistreat the interaction at the couplings of interest, so we use a four times finer thermodynamics table, start the perturbation integration ten times earlier, and sample the multipoles more finely than the default in order to remove a sporadic interpolation artefact. The cosmological parameters varied are ω_b , ω_{dm} , the angular sound horizon $100 \theta_s$, n_s , $\ln(10^{10} A_s)$ and τ , with one massive neutrino of 0.06 eV as in the Planck baseline.

2.3 Profile likelihood

For each of eight fixed values of u between 0 and 4×10^{-4} , we minimise $-\ln P$, the negative log posterior, over the six cosmological parameters and all free nuisance parameters, and form

$$\Delta\chi^2(u) = 2 \left[\min_{\theta} (-\ln P(u, \theta)) - \min_{u, \theta} (-\ln P(u, \theta)) \right]. \quad (3)$$

The only priors are Planck’s own Gaussian priors on some nuisance parameters, which are the same at every u and therefore cancel in $\Delta\chi^2$ except through the fit. Two PLIK nuisance parameters, the thermal Sunyaev–Zel’dovich–CIB correlation and the kinetic SZ amplitude, sit at their physical lower bound of zero and were pinned there, because the minimiser otherwise wasted most of its evaluations outside the allowed range. This leaves 25 free parameters for PLIK, 15 for CamSpec and 7 for PLIK-lite.

The physical coupling cannot be negative, so the null hypothesis $u = 0$ lies on the boundary of the parameter space. In that situation $\Delta\chi^2$ is distributed under the null as an equal mixture of a point mass at zero and a χ^2 with one degree of freedom (Chernoff, 1954), and the one-sided 95% upper limit is the value of u at which $\Delta\chi^2 = 2.71$. We quote this as the headline number and also give the crossing of $\Delta\chi^2 = 3.84$, which is the conventional two-sided level.

A practical difficulty is that the likelihood computed through CLASS is not a perfectly smooth function of the parameters. Changing a parameter moves the internal integration and sampling grids, which produces a small, irregular noise in $-\ln P$ of about 0.03 for PLIK and 0.1–0.2 for CamSpec. Standard trust-region minimisers stop early in such a landscape and scatter by around 0.5 in $-\ln P$ between restarts, which is too much when a $\Delta\chi^2$ precision of order 0.1 is needed. We therefore wrote a minimiser that samples the posterior at several hundred random points around the current estimate, fits a quadratic surface to them by least squares, moves to the minimum of the fitted surface within the parameter bounds and repeats until the predicted improvement is smaller than its own regression uncertainty. Averaging over many points suppresses the noise, and the regression provides both the minimum value and its statistical uncertainty; the objective

Table 1: Profile $\Delta\chi^2(u)$ for Planck 2018 low- ℓ TT and EE combined with each high- ℓ likelihood, with $f_{\text{cl}} = 1$. The CamSpec column was recomputed with the audited minimiser; a dash marks a coupling not included in that grid, and the refitted value at $u = 1.2 \times 10^{-4}$ is 2.89. The numerical uncertainty of each entry is about 0.05 for PLIK and 0.03 for the refitted CamSpec values, from repeated fits with different seeds and starting points.

u	PLIK	PLIK-lite	PR4 CamSpec
0	0	0	0
2×10^{-5}	0.55	0.56	—
5×10^{-5}	1.34	1.36	1.15
1×10^{-4}	2.71	2.76	2.42
1.5×10^{-4}	4.23	4.30	3.76
2×10^{-4}	5.94	6.04	5.35
3×10^{-4}	10.03	10.18	8.93
4×10^{-4}	15.18	15.34	—

Table 2: Upper limits on the coupling and the corresponding clump properties for $f_{\text{cl}} = 1$.

High- ℓ likelihood	u_{95}	$u(\Delta\chi^2 = 3.84)$	σ/M [$\text{cm}^2 \text{g}^{-1}$]	Σ/Q [g cm^{-2}]
PLIK (baseline)	1.00×10^{-4}	1.37×10^{-4}	$< 3.7 \times 10^{-7}$	$> 2.7 \times 10^6$
PLIK-lite	0.98×10^{-4}	1.35×10^{-4}	$< 3.7 \times 10^{-7}$	$> 2.7 \times 10^6$
PR4 CamSpec	1.12×10^{-4}	1.52×10^{-4}	$< 4.2 \times 10^{-7}$	$> 2.4 \times 10^6$
PR4 CamSpec (MCMC)	1.69×10^{-4}	—	$< 6.3 \times 10^{-7}$	$> 1.6 \times 10^6$

is also evaluated directly at the reported point as a check. The CamSpec noise has heavy tails, with a few evaluations per hundred lying four or five standard deviations from the fitted surface, so these points are excluded from the regression. The plik profiles ran on a 180-core cloud machine at roughly nine likelihood evaluations per second. The CamSpec profiles reported here were recomputed on a 16-thread workstation, at about 1.3 evaluations per second, after an audit of the minimiser described in Section 5; each point required between one and five thousand evaluations.

3 Validation

At $u = 0$ the model reduces to ΛCDM and the minimisation can be compared directly with Planck’s published best fit. For the full PLIK likelihood we obtain $\chi^2_{\text{plik}} = 2344.89$, against 2344.94 at Planck’s best fit (Planck Collaboration, 2020b), whose fit also included the lensing reconstruction. The best-fitting parameters, $\omega_b = 0.02233$, $\omega_{\text{dm}} = 0.1203$, $n_s = 0.9639$, $\tau = 0.0541$, $H_0 = 67.21 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\sigma_8 = 0.812$, agree with Planck’s to well within their uncertainties. The full and lite PLIK likelihoods give $\Delta\chi^2$ values that agree to within 1.5% at every u , which shows that pre-marginalising the foregrounds does not bias this model. Finally, a Fisher forecast made earlier in the project, with a Planck-like noise model and the six cosmological parameters free, predicted a 95% limit of $u < 1.1 \times 10^{-4}$, close to what the data give.

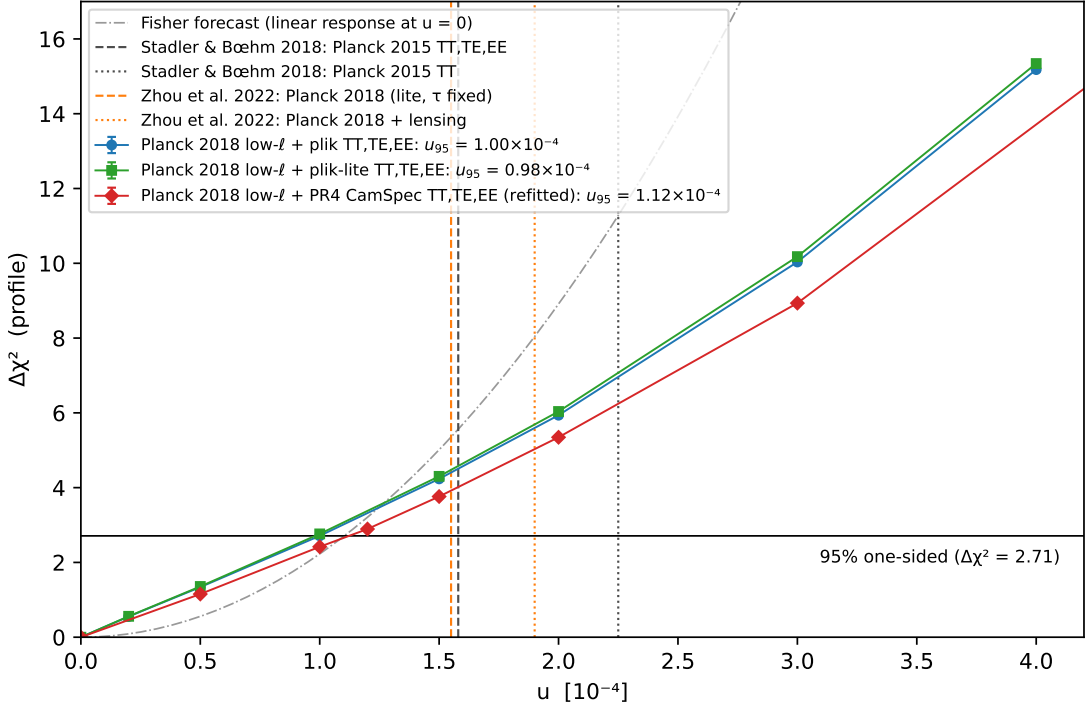


Figure 1: Profile likelihood in the clump–photon coupling u for the three likelihood combinations, with all cosmological and nuisance parameters minimised at each point. The horizontal line marks the one-sided 95% level. The vertical lines are the published marginalised 95% limits of [Stadler & Boehm \(2018\)](#) from the Planck 2015 data (grey) and of [Zhou et al. \(2022\)](#) from the Planck 2018 data without and with the lensing reconstruction (orange), and the dash-dotted curve is the Fisher forecast, which assumes that the spectra respond linearly to u about $u = 0$.

4 Results

Table 1 and Figure 1 show the profiles. In every case the best fit lies at $u = 0$ and $\Delta\chi^2$ rises steadily with the coupling. With the baseline PLIK likelihood the 95% upper limit is

$$u < 1.00 \times 10^{-4}, \quad \frac{\sigma}{M} < 3.7 \times 10^{-7} \text{ cm}^2 \text{ g}^{-1}, \quad \frac{\Sigma}{Q} > 2.7 \times 10^6 \text{ g cm}^{-2}, \quad (4)$$

and Table 2 collects the corresponding numbers for the other likelihoods. For scale, a surface density of $3 \times 10^6 \text{ g cm}^{-2}$ is that of a column of rock about ten kilometres long, so the CMB alone allows clumps that are very dense by astronomical standards but not extreme ones. Figure 2 places these limits alongside the forecasts and the published bounds.

Figure 2 includes two sets of published limits. [Stadler & Boehm \(2018\)](#) analysed the 2015 Planck release and found $u < 1.58 \times 10^{-4}$ from temperature and polarization and $u < 2.25 \times 10^{-4}$ from temperature alone. Our limit is about 35% tighter than the first of these. Part of the difference could come from the better polarization data and revised treatment of large-scale polarization in the 2018 release, but a larger part is probably statistical in origin, because their limits are Bayesian marginalised limits while ours come from a profile. When the parameter has a physical boundary at zero and the likelihood is not Gaussian in it, the two kinds of limit need not coincide, because the marginalised limit integrates the posterior mass above u , including the volume of the other parameters that remains compatible with the data, while the profile records only the best fit at each u . The most recent published limit, from [Zhou et al. \(2022\)](#), uses the same 2018 release and gives $u < 1.55 \times 10^{-4}$, also as a marginalised Bayesian limit. It is not a like-for-like comparison either, since that analysis uses the PLIK-lite likelihood, holds the optical depth fixed at $\tau = 0.054$, and varies a dark matter mass in the keV–MeV range that

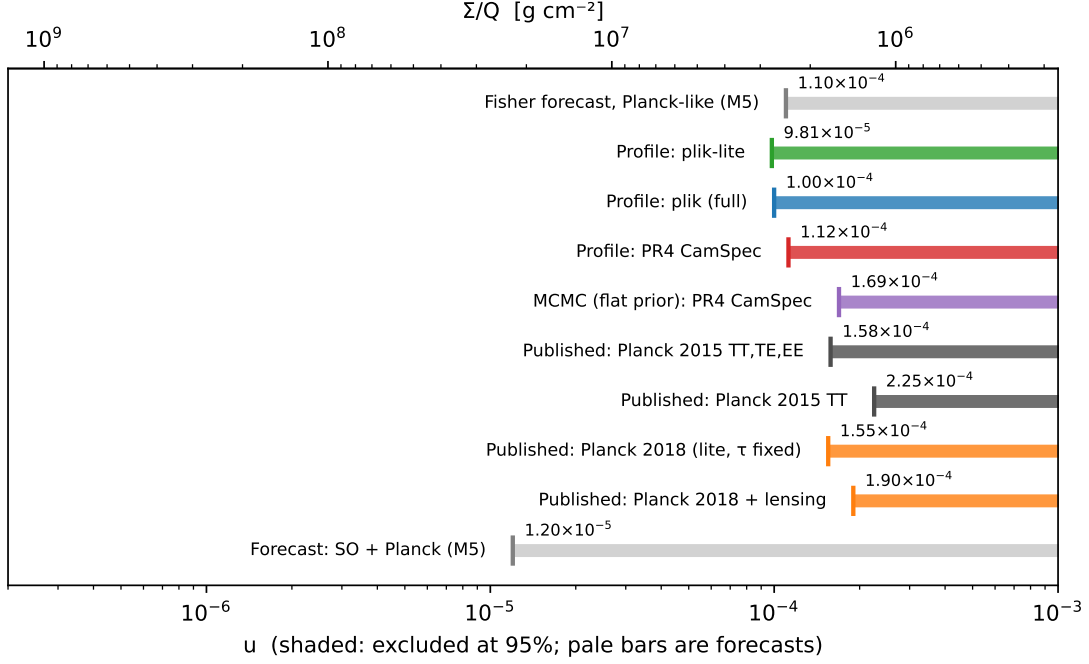


Figure 2: Summary of the 95% upper limits on u , with the equivalent minimum clump surface density on the upper axis. Shaded bars mark the excluded region. The pale bars are forecasts: the Planck-like Fisher forecast made before the data were used, and the forecast for Simons Observatory combined with Planck, which would improve the limit by nearly an order of magnitude.

gives the dark matter a small pressure, but it shows that with the same data the marginalised limit lies well above our profile limit. The Bayesian analysis in Section 7 makes the like-for-like comparison and gives $u < 1.69 \times 10^{-4}$, close to both published values.

5 How robust is the result?

Several features of the analysis give us confidence in the headline number. The pipeline reproduces Planck’s Λ CDM fit to 0.05 in χ^2 , the compressed and full PLIK likelihoods agree, the Fisher forecast agrees, and two independent high- ℓ likelihoods built from different processing of the Planck maps give limits within 12% of each other. The numerical noise in the minimisation corresponds to an uncertainty of about 0.05 in $\Delta\chi^2$ for PLIK, or roughly 2% in the limit.

The minimiser was audited after the first version of this report. Two problems were found. The first was a coordinate mismatch: sample points falling outside the prior bounds were moved onto the bounds before evaluation, while the quadratic was fitted to their original positions. The best fits lie at least five standard deviations from every bound, so this had no practical effect here; the corrected minimiser samples only inside the bounds and treats boundary minima explicitly. The second mattered more. The occasional large outliers in the CamSpec likelihood could spoil an individual regression, and a fixed convergence tolerance below the regression’s own uncertainty let the estimate wander. In the first CamSpec profile this showed up as two points whose final fits had been rejected. The CamSpec profile was therefore recomputed with outlier rejection and a convergence criterion tied to the regression error (Table 1). Repeated fits with different random seeds and an independent starting point agree to 0.008 in $-\ln P$, about 0.02 in $\Delta\chi^2$. The refitted CamSpec threshold is $(1.12 \pm 0.02) \times 10^{-4}$, 4% below the value of 1.17×10^{-4} given in the first version of this report. The PLIK profile was computed before the audit; its fits were less noisy and passed all checks, but it has not been recomputed.

The shape of the profile deserves comment. If the spectra responded linearly to u and the data were exactly those of the best-fitting Λ CDM model, $\Delta\chi^2$ would rise quadratically, as the Fisher curve in Figure 1 does. Instead it rises almost linearly from $u = 0$ and becomes gradually steeper. Two effects can produce this. The first is that the response of the spectra to u is sublinear, which was already noticed in the Fisher calculation, where the derived sensitivity changed by a few per cent with the size of the finite-difference step. The second is that the data prefer the direction opposite to the one in which u pushes the model, which is to say towards more lensing than Λ CDM predicts. The Planck 2018 spectra are known to show such a preference, with a phenomenological lensing amplitude $A_L = 1.18 \pm 0.07$ where unity is expected (Planck Collaboration, 2020b), while the PR4 CamSpec spectra show a much weaker one, $A_L \approx 1.04$ (Rosenberg et al., 2022). The initial slope of the CamSpec profile is about 20% lower than that of PLIK, which is what the second explanation would lead one to expect. If this is the explanation, the PLIK limit is somewhat tighter than the sensitivity of the data would suggest, because a statistical fluctuation towards extra lensing counts against the clumps, and the CamSpec number is the more conservative of the two. Independent support for this interpretation comes from Zhou et al. (2022), whose limit weakens from 1.55 to 1.90×10^{-4} when the Planck lensing reconstruction is added, since the reconstruction measures a lensing amplitude close to the Λ CDM value and so removes part of the pull that the power spectra exert against the coupling. A direct test would be to repeat the profile on simulated data generated from the best-fitting Λ CDM model, which isolates the sensitivity from the fluctuation, or to free A_L alongside u .

The result also rests on assumptions that are separate from the numerics. All the dark matter is taken to be in clumps. For a smaller clumped fraction only part of the matter is dragged, so the limit on u weakens, but how quickly it does so has not yet been computed with the likelihood. The Planck lensing reconstruction is not included. Lensing is the main channel through which the coupling is seen, so one might expect the reconstruction to tighten the limit, but the published analysis of Zhou et al. (2022) found the opposite, a weakening of about 20%, for the reason discussed above. We expect a similar effect here, which would move the PLIK limit towards the CamSpec value.

6 Scope of the result: what the model leaves out

The robustness checks above test the calculation within its model. A separate question is how much the answer depends on the model itself, and here the honest position is that the limit applies to a highly specialised scenario. The clumps are inserted into an otherwise standard universe, and everything else about the baryons, recombination and the initial conditions is held at its Λ CDM form. If compact baryonic clumps really made up the dark matter, they would have to form from large, early departures from homogeneity in the baryons, and it is unlikely that such a process would leave the rest of the baryonic physics untouched. The more useful question for the next stage is therefore not which arbitrary extra parameters could loosen the bound, but what additional early-universe physics necessarily accompanies the existence of the clumps, and how that physics changes the CMB.

Table 3 lists the main candidates. The first two rows have been studied in depth for reasons unrelated to dark matter. Recombination depends on the square of the density, so unresolved small-scale inhomogeneity in the baryons changes the mean ionization history even when it is invisible as a resolved perturbation. Jedamzik & Saveliev (2019) showed that primordial magnetic fields generate exactly this kind of clumping and used it to limit the field strength. Phenomenological descriptions of clumping, with a few zones of different density, have been fitted to the CMB (Rashkovetskyi et al., 2021; Thiele et al., 2021), and the effects are large enough to be tested. More recently, magnetohydrodynamic simulations coupled to recombination and Lyman- α radiative transfer have derived the ionization history from the field statistics instead of assuming a clumping law (Jedamzik et al., 2023), and fits of the resulting model to Planck,

Table 3: Assumptions fixed in the present calculation, and how a population of early compact baryonic clumps might violate them.

Ingredient	Present model	Possible departure in a clump cosmology
Diffuse baryons	smooth fluid, linear perturbations	residual small-scale clumping of the gas that did not end up in compact objects
Recombination	homogeneous, $x_e(z)$ from the mean density	density-weighted, $\langle x_e(\rho_b) \rangle$, altered by clumping and by Lyman- α transport between regions
Magnetic fields	none	primordial fields as a formation seed, with Lorentz forces and stresses
Initial conditions	adiabatic, power law	extra small-scale power or isocurvature modes from the seeding process
Clumps	pressureless, point-like, fixed σ/M , present throughout	mass and size distributions, formation at a finite redshift, evolution of σ/M
Baryon budget	ω_b from the CMB equals the BBN value	compact baryons are counted as dark matter by the CMB, in tension with BBN unless formed very early

baryon acoustic oscillation and supernova data are at least as good as those of Λ CDM while favouring a different region of parameter space (Jedamzik et al., 2025). These studies show that richer, physically coupled small-scale baryon physics can change which cosmologies fit Planck, and they also show the route by which such physics can be brought into a precision analysis. The small-scale calculation produces effective quantities such as the ionization history, and these are fed into the Boltzmann code, which is then compared with the likelihood.

Two points limit how far this freedom extends. First, the Planck data are heavily overdetermined, with temperature and polarization spectra over thousands of multipoles as well as lensing, and modifications that repair one feature commonly spoil the damping tail, the polarization phases or the matter power spectrum elsewhere; the clumping studies above found exactly this kind of restriction. Second, the coupling studied here is seen mainly as a suppression of small-scale growth and hence of lensing. Accompanying physics can only relax the limit by adding back lensing or small-scale growth, or by moving the standard parameters in a way that compensates. A modified recombination history does not add lensing, but it does change the damping tail and the sound horizon, and the Fisher analysis found the coupling to be most degenerate with θ_s and ω_b , so an interaction between the two effects through the standard parameters cannot be ruled out. Whether the net effect of a self-consistent clump cosmology is to loosen or to tighten the limit is unknown, and cannot be established without a model of how the clumps form.

For these reasons we regard the present result as the first layer of the analysis: a clean, conditional limit in which a single effect is isolated. A second layer would audit the rows of Table 3 quantitatively for a physically specified formation scenario, and a third would derive the effective inputs to the Boltzmann calculation from small-scale simulations of that scenario, in the manner of the magnetic-field work.

7 Bayesian analysis

We sampled the posterior with the Metropolis sampler in Cobaya, using the PR4 CamSpec and 2018 low- ℓ likelihoods. The coupling was given a flat prior on $[0, 10^{-3}]$. Diacoumis & Wong

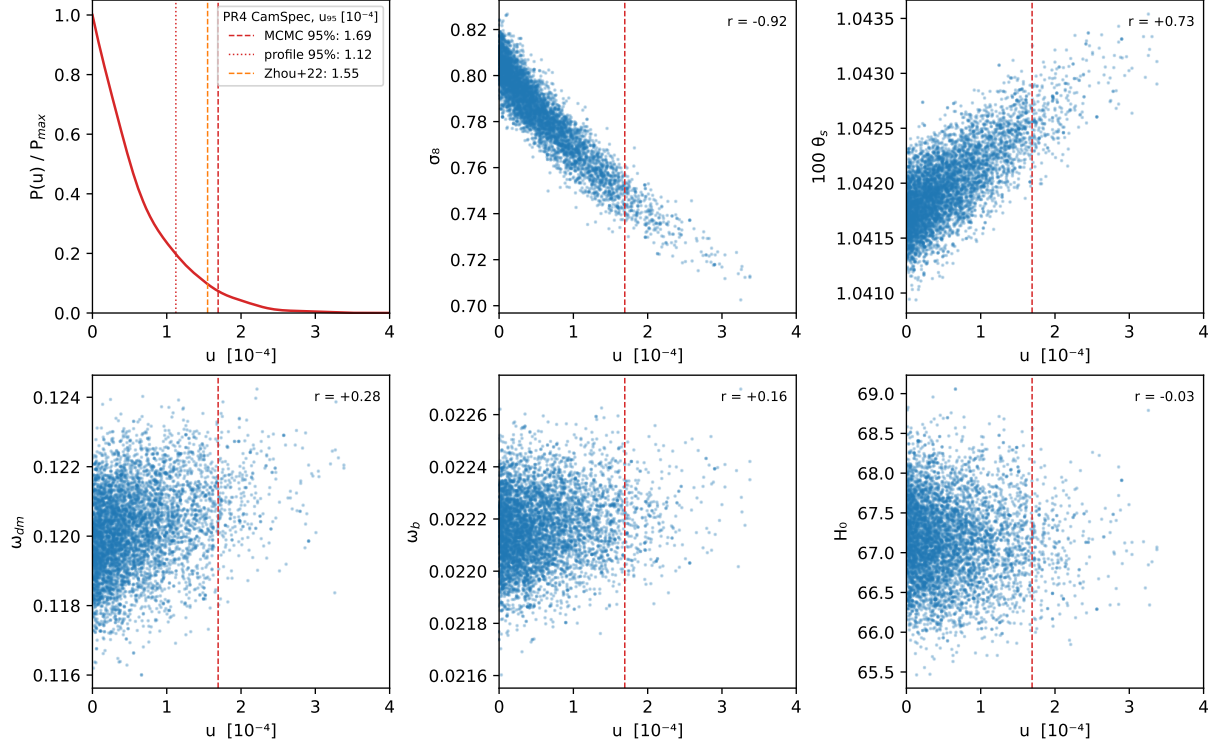


Figure 3: Markov chain Monte Carlo results for PR4 CamSpec with the 2018 low- ℓ likelihoods, $f_{\text{cl}} = 1$ and a flat prior on u . Top left: the marginalised posterior of u , with the MCMC and profile 95% limits and the published 2018 limit of Zhou et al. (2022). Other panels: posterior samples of u against the parameters it is most correlated with, with the Pearson correlation coefficient r and the MCMC 95% limit (dashed).

(2019) showed that such a prior gives limits consistent to within 20% with the Jeffreys prior, whereas a prior flat in $\log u$ makes the limit depend on an arbitrary lower cut-off. The six cosmological and nine nuisance parameters were sampled alongside u in 24 independent chains, started near the profile best fit with a proposal distribution taken from the covariance measured in the profile at $u = 0$. The sampler exploits the fact that the nuisance parameters can be varied without recomputing the CMB spectra. After removing the first 30% of each chain, the combined chains contain about 30 000 accepted samples. The Gelman–Rubin statistic for the mean of u across chains is $R - 1 = 0.02$, and the 95% limit computed from the first and second halves of every chain agrees to better than 1% (1.69 and 1.69×10^{-4}). The marginalised 95% and 99% upper limits are

$$u < 1.69 \times 10^{-4} \text{ (95\%)}, \quad u < 2.33 \times 10^{-4} \text{ (99\%)}, \quad (5)$$

corresponding to $\sigma/M < 6.3 \times 10^{-7} \text{ cm}^2 \text{ g}^{-1}$ and $\Sigma/Q > 1.6 \times 10^6 \text{ g cm}^{-2}$ at 95%. The posterior of u peaks at zero and falls monotonically (Figure 3), with no hint of a preference for a non-zero coupling. The standard parameters are close to their Λ CDM values, with $H_0 = 67.1 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $n_s = 0.962 \pm 0.004$, and $\sigma_8 = 0.78 \pm 0.02$ is lower than in Λ CDM because it is dragged down by the coupling.

The Bayesian limit is 50% weaker than the profile limit from the same likelihood. The difference is what one expects from the shape of the profile. Because $\Delta\chi^2$ rises almost linearly from $u = 0$, the likelihood falls roughly exponentially in u , and for an exponential the value that encloses 95% of the probability lies about twice as far out as the point where $\Delta\chi^2 = 2.71$. For a quadratic $\Delta\chi^2$ the ratio would be only 1.2. Integrating the profile likelihood itself, $\exp(-\Delta\chi^2/2)$, over the same flat prior gives $u < 1.96 \times 10^{-4}$. The full MCMC limit lies between that and

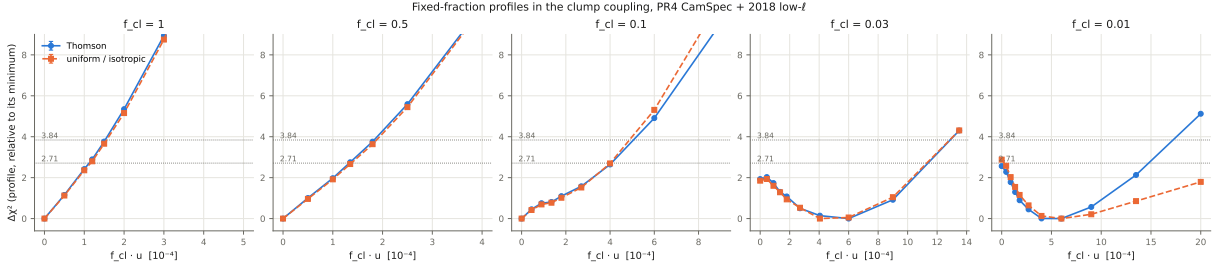


Figure 4: Fixed-fraction profiles for PR4 CamSpec and the 2018 low- ℓ likelihoods. Each panel holds the clump fraction f_{cl} fixed and shows $\Delta\chi^2$ relative to that profile’s own minimum against the product $f_{\text{cl}}u$, for the Thomson kernel (blue) and uniform, unpolarized re-emission (orange). Dotted lines mark $\Delta\chi^2 = 2.71$ and 3.84 .

the profile crossing, so marginalising over the other parameters slightly favours small couplings over what the profile alone would suggest, and there is no sign of a volume effect that inflates the Bayesian limit. The MCMC limit is close to the published Bayesian limits, 1.55×10^{-4} from the 2018 data (Zhou et al., 2022) and 1.58×10^{-4} from the 2015 data (Stadler & Boehm, 2018), which confirms that most of the apparent disagreement between our profile limit and the literature was the difference between the two kinds of statistic.

The correlations in Figure 3 show how the data see the coupling. The strongest is with σ_8 ($r = -0.92$): the clumps are dragged by the radiation and their small-scale perturbations are suppressed, so the amplitude of matter fluctuations today falls as u grows. It is this reduction in structure, seen by Planck through the lensing of the CMB, that limits u . The second is a positive correlation with the angular sound horizon θ_s ($r = +0.73$). The weaker potentials probably shift the phases of the acoustic peaks slightly, and the fit absorbs the shift by moving θ_s . That would agree with the degeneracy with θ_s found in the Fisher analysis. The correlations with ω_{dm} and ω_b are weaker ($r = +0.28$ and $+0.16$), and H_0 is essentially uncorrelated with u , so the coupling does not help with the Hubble tension. The strong anticorrelation with σ_8 also shows that the constraint rests on the amplitude of lensing, which is why the treatment of the Planck lensing excess and the addition of the lensing reconstruction matter so much to the precise value of the limit.

8 Mixtures of CDM and clumps, and the scattering law

Everything so far assumes that all of the dark matter is in clumps. We now let a fraction f_{cl} of the dark component be in clumps and the rest be collisionless CDM, $\omega_{\text{cl}} = f_{\text{cl}}\omega_{\text{dark}}$ and $\omega_{\text{CDM}} = (1 - f_{\text{cl}})\omega_{\text{dark}}$, with the total ω_{dark} fitted alongside the other parameters. The diffuse baryon density remains a separate fitted parameter, and the CDM has no photon coupling. For each of $f_{\text{cl}} = 1, 0.5, 0.1, 0.03$ and 0.01 we profile the same objective as before over all fifteen parameters, using the PR4 CamSpec and 2018 low- ℓ likelihoods, on a grid of couplings that includes $u = 0$ and was extended wherever the profile had not yet turned over and crossed the thresholds. The profiles are measured relative to their own minimum over the scanned range, which is not assumed to lie at $u = 0$, and relative to the Λ CDM fit.

We also compare two angular laws for the photon-clump interaction at equal momentum-transfer coupling. The first is Thomson scattering, which regenerates the photon quadrupole and generates polarization. The second is uniform re-emission in the clump rest frame with no polarization, the limiting case of an opaque absorber that re-radiates isotropically. Both are implemented in a patched version of CLASS that reproduces the standard code exactly for the Thomson case. In the conversion to surface density the efficiency Q is kept explicit, as before.

Before production we checked that the pipeline loads the intended build of CLASS, that

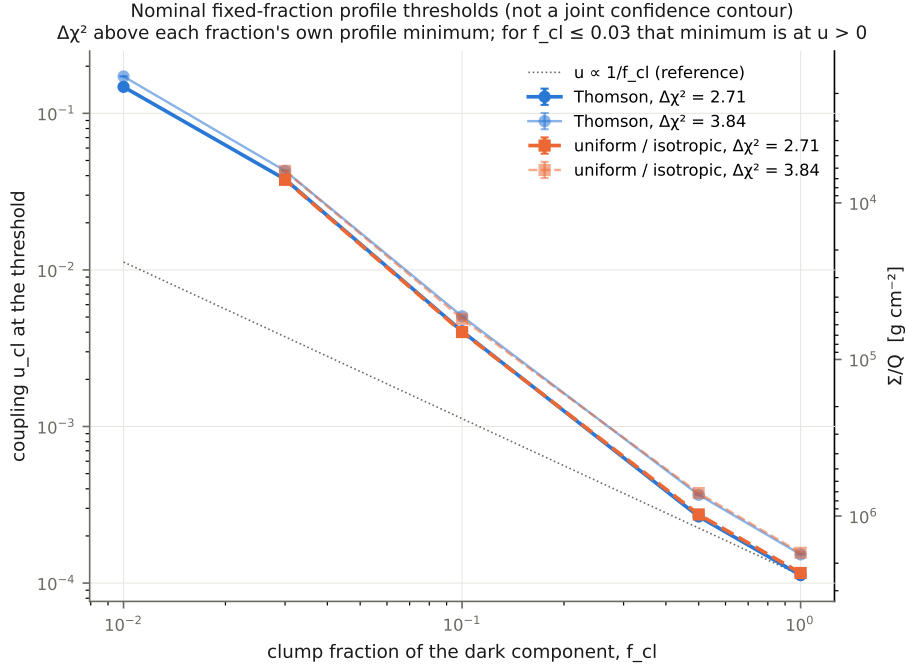


Figure 5: Coupling at which each fixed-fraction profile first reaches $\Delta\chi^2 = 2.71$ (heavy lines) and 3.84 (light lines) above its own minimum, for both kernels, with the corresponding Σ/Q on the right axis. The dotted line shows the scaling $u \propto 1/f_{\text{cl}}$ from the $f_{\text{cl}} = 1$ threshold. These are thresholds for each fraction separately and not a joint confidence region in (f_{cl}, u) . The isotropic profile at $f_{\text{cl}} = 0.01$ does not reach either level within the scanned range.

the mixture at $f_{\text{cl}} = 0$ and every fraction at $u = 0$ reproduce the Λ CDM objective within numerical noise, that both kernels share the zero-coupling limit, and that the tight-coupling approximation and the numerical precision do not affect the results. Switching the tight-coupling approximation off ten times earlier, or raising the precision of the perturbation integration and multipole sampling, changes χ^2 by at most about 0.1 relative to Λ CDM at every profile point, including the strongest couplings.

Table 4 and Figures 4 and 5 give the results. Collisionless CDM weakens the constraint on the clump coupling, and does so faster than the naive scaling $u \propto 1/f_{\text{cl}}$. Halving the clump fraction raises the threshold by a factor of 2.4, and at $f_{\text{cl}} = 0.1, 0.03$ and 0.01 the thresholds are 36, 340 and 1300 times the value for $f_{\text{cl}} = 1$, so that the product $f_{\text{cl}}u$ at the threshold grows by factors of 1.2, 3.6, 10 and 13. Two effects contribute. The drag suppresses only the clump perturbations, and after the clumps decouple the CDM pulls them back into its potential wells, so the loss of structure at a given $f_{\text{cl}}u$ is smaller when most of the dark matter is collisionless. At $f_{\text{cl}}u = 10^{-4}$, for example, σ_8 falls to 0.768 when all of the dark matter is clumped and only to 0.772 when half of it is. Below $f_{\text{cl}} \approx 0.1$ the suppression of σ_8 almost disappears ($\sigma_8 \approx 0.80$ – 0.82 along the $f_{\text{cl}} = 0.01$ profile) and the profile is shaped instead by the effect of a small, strongly coupled component on the damping tail. In terms of the clumps themselves, the minimum surface density falls from $2.4 \times 10^6 \text{ g cm}^{-2}$ for $f_{\text{cl}} = 1$ to 1.0×10^6 , 7×10^4 , 7×10^3 and $2 \times 10^3 \text{ g cm}^{-2}$ for the smaller fractions.

At the two smallest fractions the profile does not rise from $u = 0$. It first falls below Λ CDM, reaching $\Delta\chi^2 \approx -1.9$ at $f_{\text{cl}} = 0.03$ and -2.6 at $f_{\text{cl}} = 0.01$ (Thomson), before rising steeply. The improvement comes from the low- ℓ temperature likelihood, with a smaller contribution from the high- ℓ likelihood, and is accompanied by a steady rise of the fitted spectral index from 0.961 to about 0.97. A higher n_s lowers the predicted large-scale power, which the known deficit in the Planck low- ℓ temperature spectrum favours, and the modification of the damping tail by

Table 4: Nominal fixed-fraction profile thresholds ($\Delta\chi^2$ above each profile’s own minimum) and the corresponding minimum surface density for the Thomson kernel. The uncertainties are numerical only and do not include the error of interpolating between grid points, which is a few per cent; the last column gives the depth of the profile minimum relative to Λ CDM.

f_{cl}	u at $\Delta\chi^2 = 2.71$		u at 3.84	Σ/Q [g cm^{-2}]	u at min.	min. – Λ CDM
	Thomson	isotropic	Thomson	Thomson	Thomson	Thomson
1	$(1.12 \pm 0.02) \times 10^{-4}$	$(1.16 \pm 0.02) \times 10^{-4}$	1.52×10^{-4}	2.4×10^6	0	+0.02
0.5	$(2.66 \pm 0.04) \times 10^{-4}$	$(2.74 \pm 0.04) \times 10^{-4}$	3.66×10^{-4}	1.0×10^6	0	+0.03
0.1	$(4.06 \pm 0.04) \times 10^{-3}$	$(4.01 \pm 0.03) \times 10^{-3}$	5.06×10^{-3}	6.6×10^4	0	+0.02
0.03	$(3.79 \pm 0.02) \times 10^{-2}$	$(3.77 \pm 0.02) \times 10^{-2}$	4.30×10^{-2}	7.1×10^3	0.02	−1.91
0.01	$(1.48 \pm 0.01) \times 10^{-1}$	not reached by 0.2	1.72×10^{-1}	1.8×10^3	0.06	−2.56

the coupled clumps allows the high- ℓ fit to accommodate it. This is a degeneracy of the model with a well-known feature of the data. An improvement of two to three in χ^2 , obtained with two additional parameters after a search, is not evidence for clumps, and other one-parameter extensions such as a running spectral index exploit the same feature. A meaningful test would compare the clump model with such alternatives and add data that do not share the Planck low- ℓ feature, such as the lensing reconstruction or ground-based small-scale spectra.

The scattering law changes these conclusions very little. Wherever both thresholds are reliable, the isotropic kernel moves them by +3% at $f_{\text{cl}} = 1$ and 0.5 and by about −1% at $f_{\text{cl}} = 0.1$ and 0.03, comparable to the numerical uncertainty. Below the thresholds the two profiles differ by at most 0.2 in $\Delta\chi^2$, the isotropic kernel fitting slightly better because the clumps then produce no polarization of their own. The exception is the smallest fraction at strong coupling. At $f_{\text{cl}} = 0.01$ and $u \gtrsim 0.1$ the clumps contribute an appreciable share of the scattering after recombination, and Thomson scattering off them generates polarization that the high- ℓ data disfavour. The Thomson profile therefore turns up sharply and crosses the threshold at $u = 0.15$, while the isotropic profile is still only 1.8 above its minimum at $u = 0.2$, the largest coupling scanned. No upper limit is claimed for the isotropic kernel at $f_{\text{cl}} = 0.01$; the calculation itself remains valid beyond this range, but the profile was not extended further. This difference persists when the tight-coupling treatment or the numerical precision is changed, so it is a property of the models and not of the approximations used to compute them.

9 Summary

If all of the dark matter were in compact clumps coupled to the photons through a geometric cross-section, and nothing else departed from Λ CDM, the Planck 2018 temperature and polarization spectra require the coupling to satisfy $u < 1.0 \times 10^{-4}$ at 95% confidence, or equivalently a clump surface density $\Sigma/Q > 2.7 \times 10^6 \text{ g cm}^{-2}$. The final Planck processing gives a limit 12% weaker, and the difference between the two likelihoods is consistent with the known difference in their preference for extra lensing. The limit agrees with an advance forecast and is tighter than the published marginalised limits from the 2015 and 2018 data, as a profile limit on a parameter with a physical boundary is expected to be. The application of the particle bound to macroscopic dark matter was first made by [Jacobs et al. \(2015\)](#); what this analysis adds is a dedicated treatment of opaque clumps with current likelihoods. The limit is conditional on that narrow model, and physics that plausibly accompanies clump formation, notably small-scale inhomogeneity of the remaining gas and its effect on recombination, is not included. A Bayesian analysis with the PR4 likelihood gives $u < 1.69 \times 10^{-4}$, or $\Sigma/Q > 1.6 \times 10^6 \text{ g cm}^{-2}$. The difference from the profile limit follows from the nearly linear shape of the profile, and brings the result into line with the published Bayesian limits. When only a fraction of the dark matter is clumped and the rest is collisionless CDM, the permitted coupling grows faster than the inverse

of that fraction, reaching $u \approx 4 \times 10^{-3}$ at $f_{\text{cl}} = 0.1$ and 0.15 at $f_{\text{cl}} = 0.01$, with minimum surface densities of 7×10^4 and $2 \times 10^3 \text{ g cm}^{-2}$. Whether the clumps scatter like electrons or re-emit isotropically changes these numbers by a few per cent, except for the smallest fraction at strong coupling. At small fractions the profiles dip slightly below ΛCDM , a degeneracy with the Planck low- ℓ deficit that is not significant. The main open items are a test of how much the Planck lensing excess tightens the bound, a comparison of the small-fraction dip with other extensions that exploit the same feature of the data, and an audit of which standard assumptions a specific formation scenario would violate.

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